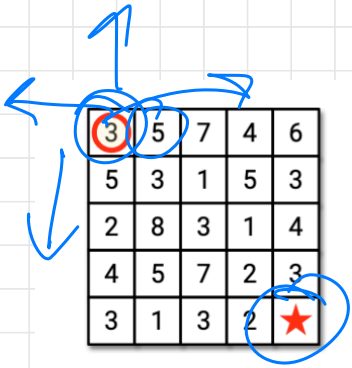
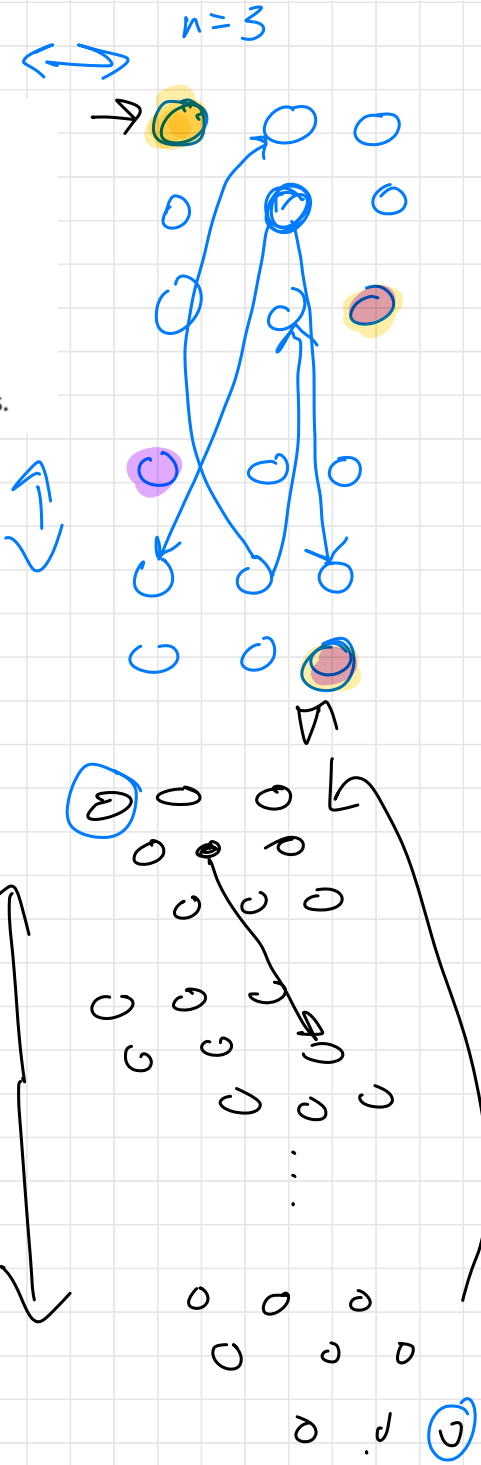


3	5	7	4	6
5	3	1	5	3
2	8	3	1	4
4	5	7	2	3
3	1	3	2	★



3	5	7	4	6
5	3	1	5	3
2	8	3	1	4
4	5	7	2	3
3	1	3	2	★

A 5 × 5 number maze that can be solved in eight moves.



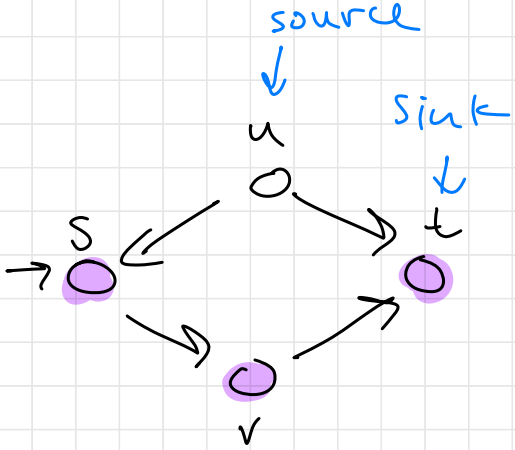
Directed $v \to w \neq w \to v$

tail: v, head: w

$O(V+E)$

WHATEVERFIRSTSEARCH(s):

```
put s into the bag
while the bag is not empty
    take v from the bag
    if v is unmarked
        mark v
        for each edge v→w
            put w into the bag
```



call u a source if no incoming edges

call v a sink if v has no outgoing edges

u can reach v $\Rightarrow$ there is a directed path from u to v

s can reach t

t cannot reach s

u, v are strongly connected if u can reach v and v can reach u

runtime to check u, v strongly connected?

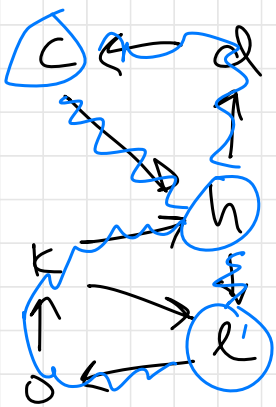
run WFS(u): did v get marked?

run WFS(v): did u get marked?

$O(V+E)$

we say G is strongly connected if all pairs of nodes can reach one another.

Give an algorithm to determine whether directed graph G is strongly connected.



(1) naively (don't worry abt runtime)

WFS(v)

WFS on all other vertices, too - do they reach v? $O(V(V+E))$

(2) smart $O(V+E)$

hint: run WFS twice

modify graph

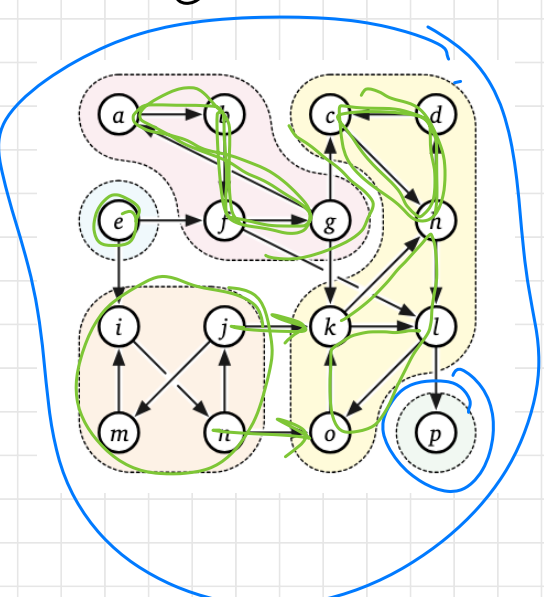
select some v

WFS(v)

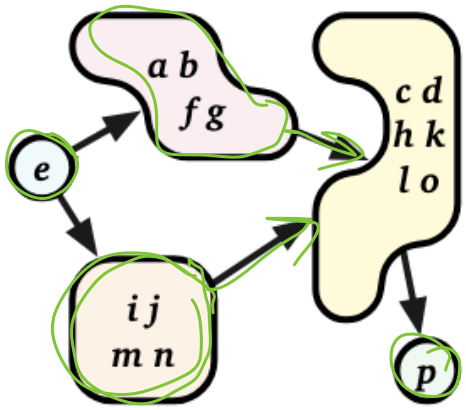
reverse all edges of G

WFS(v)

↳ all nodes that can reach v



p cannot reach e, so G is not S.C.



$O(V+E)$

metagraph